

Probabilistic graphical models

1 Introduction

2 Joint and Conditional Probability

2.1 Joint probability

For random variables X and Y , the *joint probability* $P(X, Y)$ assigns a probability to every combination of outcomes of X and Y simultaneously. For discrete variables it is a table summing to 1:

$$\sum_x \sum_y P(X = x, Y = y) = 1.$$

The joint distribution over all variables in a model is the most information one can have: any other quantity (marginal, conditional) can be derived from it by summation or division. This is also why representing it directly (as a table) is intractable for many variables — a graphical model exists precisely to represent the joint compactly using structure (independence).

2.2 Marginal probability

The *marginal* probability of X is obtained by summing (or integrating, for continuous variables) the joint over the other variable(s):

$$P(X = x) = \sum_y P(X = x, Y = y).$$

This is often called *summing out* or *marginalizing out* Y .

2.3 Conditional probability

The *conditional probability* of Y given X is the probability of Y restricted to the world where $X = x$ is known, renormalized to sum to 1:

$$P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)}, \quad \text{provided } P(X = x) > 0.$$

Rearranging gives the **chain rule (product rule)**:

$$P(X, Y) = P(Y \mid X) P(X) = P(X \mid Y) P(Y).$$

For n variables, repeated application gives the general chain rule:

$$P(X_1, \dots, X_n) = P(X_1) P(X_2 \mid X_1) P(X_3 \mid X_1, X_2) \cdots P(X_n \mid X_1, \dots, X_{n-1}),$$

which holds for *any* ordering of the variables. This is exactly the factorization a Bayesian network exploits: it fixes an ordering (a topological order of the DAG) and assumes each X_i only needs its parents, not all predecessors, i.e. $P(X_i | X_1, \dots, X_{i-1}) = P(X_i | \text{Parents}(X_i))$.

2.4 Bayes' rule

Equating the two chain-rule expressions for $P(X, Y)$ gives **Bayes' rule**:

$$P(X | Y) = \frac{P(Y | X) P(X)}{P(Y)}.$$

This lets us "invert" a conditional: given how likely evidence Y is under each hypothesis X (the likelihood $P(Y | X)$) and a prior $P(X)$, we can compute the posterior $P(X | Y)$.

2.5 Independence

X and Y are *independent*, written $X \perp Y$, if knowing one tells you nothing about the other:

$$P(X, Y) = P(X) P(Y) \iff P(X | Y) = P(X).$$

More generally, X and Y are *conditionally independent* given Z , written $X \perp Y | Z$, if

$$P(X, Y | Z) = P(X | Z) P(Y | Z) \iff P(X | Y, Z) = P(X | Z).$$

Conditional independence is the key assumption graphical models encode structurally (via d-separation in a Bayesian network, or separation in a Markov network): it is what turns the full joint chain rule above into a compact product over local factors, such as the noisy-OR and linear Gaussian CPDs discussed next.

3 The Noisy-OR Model

3.1 Motivation

In a Bayesian network, a node Y with k binary parents $X_1, \dots, X_k$ needs a conditional probability table (CPT) specifying $P(Y | X_1, \dots, X_k)$ for all 2^k parent configurations. This grows exponentially in the number of parents, which is both hard to elicit from experts and expensive to store.

The noisy-OR model is a canonical example of a *compact* (parametric) CPT: it assumes each parent X_i independently has some chance of *causing* $Y = 1$, and Y is on if *any* of the causes succeed. This reduces the number of parameters from $O(2^k)$ to $O(k)$.

3.2 Setup

Each parent X_i is a binary cause of the binary effect Y . Introduce independent "inhibitor" variables: even when $X_i = 1$, the cause is inhibited (fails to trigger Y) with probability q_i . Equivalently, X_i succeeds in causing Y with probability $1 - q_i$, independently of the other causes.

$Y = 0$ iff every active parent's cause is inhibited. So

$$P(Y = 0 | X_1, \dots, X_k) = \prod_{i: X_i=1} q_i, \quad P(Y = 1 | X_1, \dots, X_k) = 1 - \prod_{i: X_i=1} q_i.$$

Only k parameters $q_1, \dots, q_k$ are needed, one per parent.

3.3 Leak probability

In practice Y can sometimes be true even when all listed parents are 0 (an unmodeled or background cause). This is handled with a *leak* parameter $q_0 \in (0, 1]$, treated as an always-active virtual parent:

$$P(Y = 0 \mid X_1, \dots, X_k) = q_0 \prod_{i: X_i=1} q_i.$$

3.4 Example

Suppose Y = “fever” has parents X_1 = flu, X_2 = malaria, X_3 = cold, with inhibition probabilities $q_1 = 0.4$, $q_2 = 0.1$, $q_3 = 0.7$ (so flu causes fever with probability 0.6, malaria with 0.9, cold with 0.3). If a patient has flu and a cold but not malaria ($X_1 = 1, X_2 = 0, X_3 = 1$):

$$P(Y = 0 \mid X_1 = 1, X_2 = 0, X_3 = 1) = q_1 q_3 = 0.4 \times 0.7 = 0.28, \quad P(Y = 1 \mid \cdot) = 0.72.$$

3.5 Properties

- **Monotonic:** adding an active cause can only increase $P(Y = 1)$, since it multiplies another factor $q_i \in (0, 1]$ into $P(Y = 0)$.
- **Symmetric in the causes:** the causes contribute independently; order doesn’t matter, only which subset is active.
- **Special case:** if every $q_i \rightarrow 0$, this reduces to the deterministic logical OR of the parents.
- **Generalizations:** noisy-AND (dual model, symmetric using De Morgan’s law) and noisy-MAX (extension to multi-valued, non-binary variables).

4 The Linear Gaussian Model

4.1 Motivation

Noisy-OR gives a compact CPD for a binary effect with binary causes. When variables are instead continuous, a CPT is not even an option (there are infinitely many parent configurations), so we need a parametric form for $P(Y \mid X_1, \dots, X_k)$. The linear Gaussian model is the standard compact CPD for continuous variables in a Bayesian network.

4.2 Setup

Let Y be a continuous child with continuous parents $X_1, \dots, X_k$. The linear Gaussian model assumes Y is a noisy linear combination of its parents:

$$Y = \beta_0 + \sum_{i=1}^k \beta_i X_i + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2),$$

so that

$$P(Y \mid X_1, \dots, X_k) = \mathcal{N}\left(\beta_0 + \sum_{i=1}^k \beta_i X_i, \sigma^2\right).$$

Only $k + 2$ parameters are needed: the weights $\beta_0, \dots, \beta_k$ and the noise variance σ^2 , instead of a distribution for every parent value.

4.3 Key property: joint Gaussian

If every variable in the network is linear Gaussian (each node's CPD has this form given its parents), then the joint distribution over all variables is a multivariate Gaussian, and every marginal and every conditional (given evidence) is Gaussian as well. This is what makes exact inference in Gaussian Bayesian networks tractable in closed form (via the mean vector and covariance matrix), analogous to how tabular CPDs make exact inference tractable for discrete networks.

4.4 Example

Let Y = “exam score” depend linearly on X_1 = hours studied and X_2 = hours slept, with $\beta_0 = 50, \beta_1 = 4, \beta_2 = 2, \sigma^2 = 25$. For a student with $X_1 = 5, X_2 = 7$:

$$E[Y \mid X_1 = 5, X_2 = 7] = 50 + 4(5) + 2(7) = 84, \quad Y \mid X_1 = 5, X_2 = 7 \sim \mathcal{N}(84, 25).$$

4.5 Properties

- **Linear in the mean:** the conditional mean is an affine function of the parents; the variance σ^2 does not depend on the parents' values (homoscedastic noise).
- **Closed under the network:** linear Gaussian CPDs throughout a network give a jointly Gaussian distribution, so inference reduces to linear algebra rather than integration.
- **Conditional linear Gaussian (CLG):** a common extension mixes discrete and continuous parents, using a discrete parent to select among different sets of (β, σ^2) parameters (a different linear model per discrete configuration).
- **Contrast with noisy-OR:** both are $O(k)$ -parameter compact CPDs, but noisy-OR targets binary cause/effect variables while linear Gaussian targets continuous ones.

By discretizing our random variables, we can use the noisy-or model however we can lose information. By discretizing our random variables, we can use the linear Gaussian model however we can lose information.